LIE-DSM: Leveraging Single Remote Sensing Imagery to Enhance Digital Surface Model Resolution

Siyuan Zhang, Jingxian Dong, Twaha B. Kabika, Hongsen Cai, Aliaksandr Chervan, Xin Li, and Wenguang Hou

***Abstract*— Digital surface models (DSMs) have numerous valuable applications in infrastructure and industrial development. However, the spatial resolutions of DSMs are often limited due to data acquisition constraints, resulting in potential inaccuracies in these applications. Recently, deep learning-based super-resolution algorithms have been utilized to enhance the accuracy of DSMs. Despite their success, these algorithms still possess limited accuracy and robustness. As such, a new DSM super-resolution algorithm named LIE-DSM is proposed, which leverages single remote sensing imagery (RSI) to enhance the accuracy of low-resolution (LR) DSM. Specifically, we introduce a dual-input neural network tailored to the characteristics of RSIs and DSMs to generate a high-resolution (HR) output. Experiments show that LIE-DSM attains outstanding performance in all metrics and the improvements exceeds 15% in high-ratio upsampling. Moreover, the visualized results showcase more accurate shapes, crisper edges, and a distribution closely resembling the ground truth.**

***Index Terms*— Digital elevation model (DEM), digital surface model (DSM), dual-input network, super resolution.**

I. Introduction

AN ACCURATE depiction and representation of the Earth's surface in three dimensions is crucial in various fields of Earth sciences [1], [2]. The widely used models are digital surface models (DSMs) and digital elevation models (DEMs). Typically, acquiring high-precision DSMs requires multiview high-resolution (HR) remote sensing images or light detection and ranging (LiDAR) technology. However, these approaches incur significant expense in both office operations and fieldwork costs [3]. To reduce costs, super-resolution algorithms originally designed for natural images are employed on DSM/DEM data [4]. This is feasible because the grid DSM can be treated as a single-channel image, where the planar coordinates correspond to the pixel positions and the elevation values represent the pixel values.

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The dominant image super-resolution algorithms [6], [7], [8], [9] focus on establishing a mapping between low-resolution (LR) inputs and the desired HR outputs based on deep learning techniques. Drawing inspiration from these algorithms, some researchers [10], [11], [12] have extended the application of state-of-the-art (SOTA) natural image super-resolution networks to DSM/DEM data, incorporating remote sensing priors or fine-tuning operators. Despite their success, they still pose the limited accuracy. The primary reason is that the super-resolution model is inherently ill-posed, as multiple HR data can correspond to the same LR degradation. In contrast to natural images, obtaining HR DSM datasets is significantly limited. Moreover, the ill-posed nature is further exacerbated when the upsampling factor is increased. Consequently, it is difficult to generate satisfying HR results by only relying on the LR DSM.

Considering the limitations mentioned above, incorporating additional information into the DSM super-resolution is necessary. The inexpensive and continuous contours have been utilized to provide guidance information for DEM super-resolution [13]. In our opinion, the higher resolution remote sensing imagery (RSI) can serve as more valuable option. RSI and DSM can be regarded as different modalities of the same scene, with DSM representing geometry and RSI providing a mixture of geometry, materials, and lighting conditions. Regarding DSM super-resolution, LR DSM offers global structural information but lacks high-frequency details, whereas the higher resolution RSI can supplement it. Moreover, obtaining the HR RSI is generally easier than obtaining a DSM with the same resolution, making it valuable for realistic applications.

Specifically, we introduce a novel algorithm for DSM super-resolution, named as LIE-DSM. It takes LR DSM and HR RSI as inputs and outputs the desired HR DSM in an end-to-end manner. Considering the high complexity of RSIs and the dispersion of DSMs, the network architecture is well-designed with specific functional modules to achieve remarkable performance. Besides, we propose a shadow-weighted loss to mitigate the impact of shadow areas on DSM super-resolution.[1]

To summarize, our contributions are as follows.

1) We introduce a novel algorithm (LIE-DSM) which leverages the corresponding HR RSI to enhance the resolution of DSM. To the best of our knowledge,

[1]The test code is available at: https://zenodo.org/records/10199223

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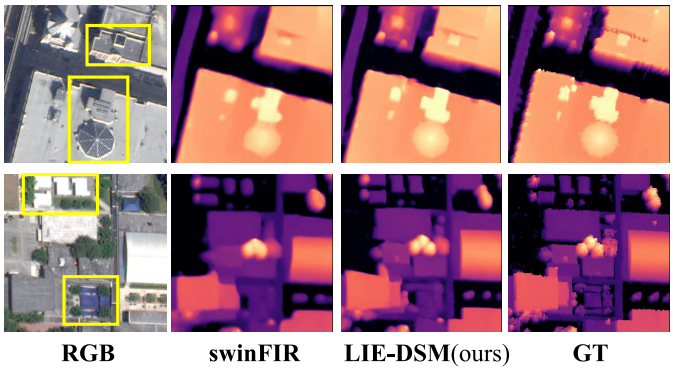


Fig. 1. Visual comparison of the proposed LIE-DSR with swinFIR [5]. Our approach achieves better performance on dense buildings and building edges.

no existing algorithms in the DSM super-resolution field resemble ours.

2) Considering the characteristics of inputs, we propose a network with self-designed functional modules and a shadow-weighted loss function to achieve remarkable performance.
3) The experimental results demonstrate that our algorithm improves remarkably in metrics, particularly under the 16× scaling factor, where our method achieves improvements exceeding 15%. Moreover, the visualized results showcase more accurate shapes, crisper edges and a closer distribution to the ground truth. (Fig. 1 exhibits two samples.)

II. Related Works

A. Conventional Super-Resolution Methods

Early approaches to enhance DEM/DSM resolution relied on interpolation-based techniques, including linear interpolation and kriging interpolation [14], [15], [16]. While these methods offered computational efficiency, they failed to capture intricate details, resulting in unsatisfactory results [17]. Subsequently, patch-based super-resolution methods [18], [19], [20] gained attention as an alternative. These methods utilized techniques such as representation [21] and nonlocal means [22] which have been adapted to exploit local image patches for reconstruction.

These methods have shown promising results over traditional interpolation approaches, making them suitable candidates for various geospatial applications. However, challenges like edge preservation, oversmoothing, and generalization to different terrains limit their applicability in situations where high accuracy, efficiency, and robustness across diverse geographies are essential [23].

B. Learning-Based Super-Resolution Methods

With the advancement of machine learning, super-resolution approaches based on deep learning have become the mainstream. These methods can be categorized into two types based on their data source, direct super-resolution and indirect super-resolution methods.

1) *Direct Methods:* Aim to establish a mapping between LR and HR samples, inspired by natural image super-resolution algorithms [24], [25], [26]. Chen et al. [4] designed a three layers CNN model, where they input LR DEM in the first layer to detect some features from the input, then integrate the detected features into some compressed ones, and finally transform the compressed features into the desired HR DEM. Xu et al. [10] employed transfer learning by pretraining on natural images to enhance network performance and robustness. Subsequently, alterations have been proposed, such as RSPCN [27], TfaSR [28], and GISR [29], which focus more on terrain features. Despite their success, both methods have limitations in recovering damaged structural information from LR data and distinguishing between multiple HR sources with the same LR DSM.
2) *Indirect Methods:* Employ super-resolution algorithms to enhance the resolution of multiview RSIs first. Then, these enhanced data are utilized to generate higher resolution DSM through stereo-matching and triangulation. Zhang et al. [30] initially proposed this approach. However, some experiments [31], [32] revealed that it may show errors in object boundaries and texture-deficient regions. Furthermore, in practical applications, the indirect algorithms pose drawbacks in terms of their complexity and automation due to the inclusion of stereo-matching.

To overcome these challenges, our LIE-DSM takes LR DSM and HR RSI as dual inputs, and outputs the desired HR DSM. By leveraging a corresponding higher resolution RSI as guidance, plenty of fine-grained geometric information can be extracted to enhance the DSM super-resolution process. This approach significantly alleviates the ill-posed nature of the problem, leading to better performance and superior stability.

III. Methodology

A. Problem Analysis

DSMs are crucial in providing valuable elevation data and capturing the geometric structure, particularly in urban areas. However, densely arranged artificial structures with diverse shapes poses a challenge to the continuity of DSM data. As a result, DSMs exhibit more discrete data distributions than DEMs. This discreteness necessitates effectively utilizing of high-frequency information from RSI, which is essential for achieving accurate DSM super-resolution. Another issue is the potential adverse effects caused by shadow areas in RSI. Shadows arise from the interplay between geometry and lighting conditions. Strong and fragmented shadows can obscure original information, blur object boundaries, and complicate the differentiation between neighboring objects.

Building upon the analysis mentioned above, the solution should focus on enhancing performance in the following aspects: extracting high-frequency information from RSI, integrating features from both RSI and DSM, and mitigating the negative impacts of shadow areas in RSI. For the first two aspects, we have designed specific functional blocks within our network architecture. Additionally, to address issues related to shadows, a shadow-weighted loss function is proposed. These approaches enable our LIE-DSM model to exhibit highly task-specific characteristics, optimized for DSM super-resolution.

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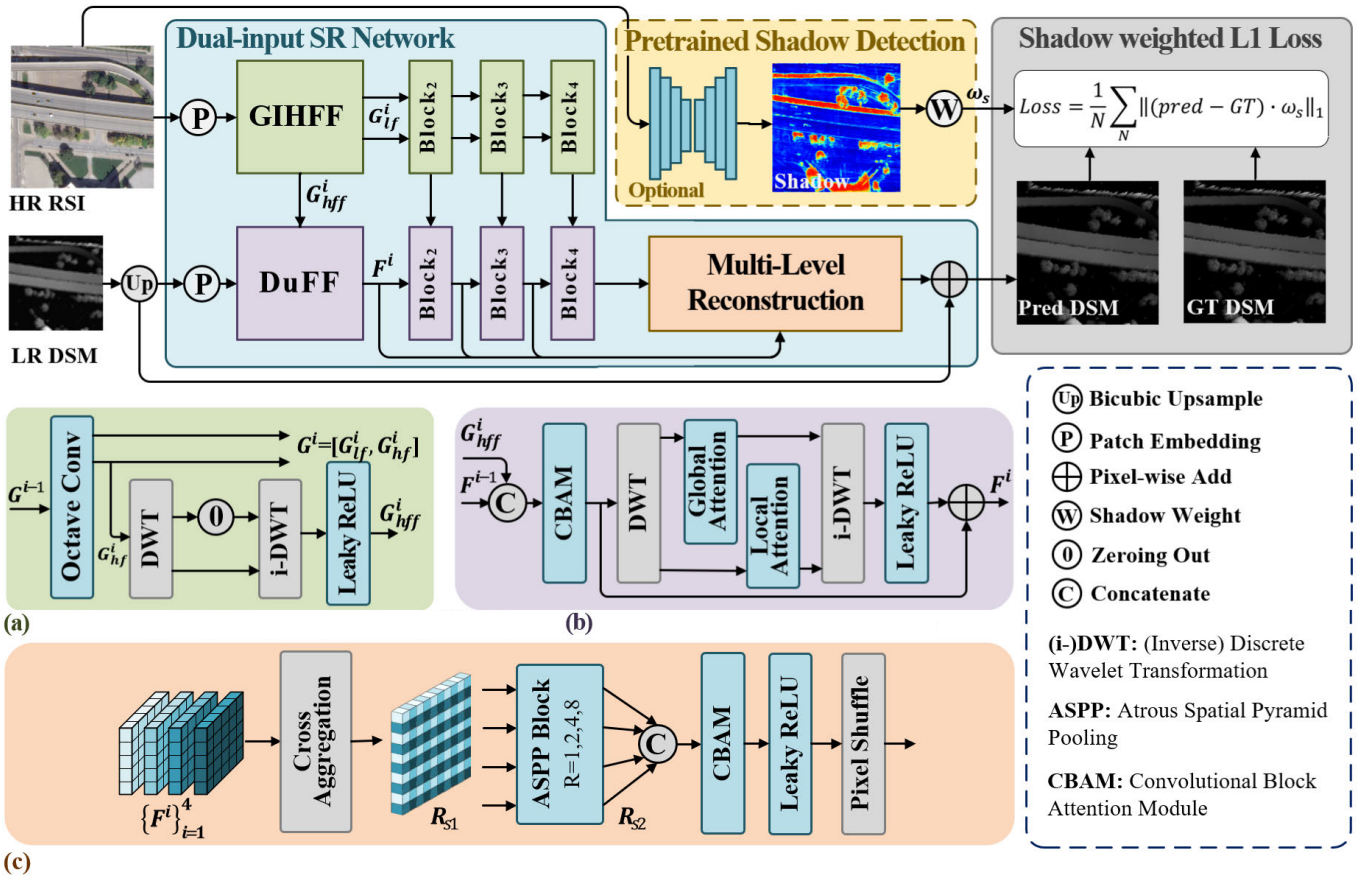


Fig. 2. Architecture of our network. It takes HR RSI and LR DSM as inputs, and outputs the desired HR DSM in an end-to-end manner. Additionally, a shadow-weighted loss is proposed to train the model. (a) GIHFF. (b) DuFF. (c) MLR.

B. Network Architecture

1) Overall Structure: The architecture of our network is illustrated in Fig. 2. It takes the LR DSM (D_{LR}) and the corresponding HR RSI (I_{HR}) as inputs and generates the desired HR DSM (D_{SR}) in an end-to-end manner. Specifically, D_{LR} is enhanced to match the resolution of the I_{HR} by bicubic interpolation $\mathcal{F}_{\mathrm{UP}}(\cdot)$. The patch embedding module $\mathcal{F}_{\mathrm{PE}}(\cdot)$ generates the initial RSI features G^0 and the initial fused features F^0. Then, a series of guided-image high-frequency feature (GIHFF) blocks and dual-input feature fusion (DuFF) blocks are utilized to perform the deep feature extraction. The GIHFF generates the filtered high-frequency RSI features G^i_{hff}, while the DuFF provides the fused features F^i by combining G^i_{hff} with the previous level F^{i-1}. Finally, the multilevel reconstruction (MLR) block simultaneously takes $\{F^i\}_{i=1}^4$ for reconstruction. The predicted result is obtained by adding the MLR output to $\mathcal{F}_{\mathrm{UP}}(D_{\mathrm{LR}})$.

2) GIHFF Block: The high-frequency information in I_{HR} plays a crucial role in accurately reconstructing the missing details of D_{LR}. Conversely, the low-frequency information in I_{HR} is less relevant and can introduce undesirable noise. To mitigate this, we propose the GIHFF module, as shown in Fig. 2(a). GIHFF receives the previous level G^{i-1}, outputs RSI feature G^i and the filtered high-frequency RSI feature G^i_{hff}. The formula is as follows:

$$G^i, \quad G^i_{\mathrm{hff}} = \mathcal{F}_{\mathrm{GIHFF}}(G^{i-1}). \tag{1}$$

We first borrow octave convolution [33] to adaptively separate the low-frequency G^i_{lf} and high-frequency G^i_{hf} as the next-level G^i. To obtain a purer high-frequency feature for the fusion block, the G^i_{hf} is further filtered by wavelet transformation to produce G^i_{hff}, following these equations:

$$[cA, cH, cV, cD]^i = \mathcal{F}_{\mathrm{DWT}}\left(G^i_{\mathrm{hf}}\right) \tag{2}$$

$$G^i_{\mathrm{hff}} = \mathcal{F}_{\mathrm{LReLU}}(\mathcal{F}_{\mathrm{iDWT}}([0, cH, cV, cD]^i)). \tag{3}$$

3) DuFF Block: The detailed architecture of the DuFF block is shown in Fig. 2(b). It takes G^i_{hff} and the previous level feature F^{i-1} as inputs, producing the current level fused feature F^i in the following equation:

$$F^i = \mathcal{F}_{\mathrm{DuFF}}\left(G^i_{\mathrm{hff}}, F^{i-1}\right). \tag{4}$$

The two sources are initially concatenated, followed by the CBAM [34], to obtain the initial fusion feature F^i_{s1}. Then, wavelet transformation is applied to split F^i_{s1} into different frequency components. The high-frequency components $[cH, cV, cD]$ undergo local attention to emphasize important local patterns, while the low-frequency components $[cA]$ undergo global attention for a broader contextual understanding. After obtaining the further fusion feature F^i_{s2} by inverse transformation, combining the two features can get the final fused feature F^i. These formulas are as

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follows:

$$[cA, cH, cV, cD]^i = \mathcal{F}_{\text{DWT}}\left(F_{s1}^i\right) \tag{5}$$

$$F_{s2}^i = \mathcal{F}_{\text{iDWT}}(\mathcal{F}_{\text{GA}}(cA), \mathcal{F}_{\text{LA}}(cH, cV, cD)) \tag{6}$$

$$F^i = \mathcal{F}_{\text{LReLU}}(F_{s2}^i) + F_{s1}^i. \tag{7}$$

4) MLR Block: In the context of super-resolution tasks, the rich, detailed information in shallow features is also crucial in addition to high-level semantic features. We specifically design the MLR block, as shown in Fig. 2(c). This module takes in multilevel fused features $\{F^i\}_{i=1}^4$ simultaneously and aims to reconstruct the residual between D_{SR} and $\mathcal{F}_{\text{UP}}(D_{\text{LR}})$, as follows:

$$D_{\text{SR}} = \mathcal{F}_{\text{MLR}}\left(\{F^i\}_{i=1}^4\right) + \mathcal{F}_{\text{UP}}(D_{\text{LR}}). \tag{8}$$

Since $\{F^i\}_{i=1}^4$ maintains the same spatial resolution, we employ a cross-fusion approach to concatenate the features, resulting in an initial reconstructed feature R_{s1}. This approach brings the features closer in position, enhancing feature correlation while achieving an initial resolution enhancement. Furthermore, to expand the receptive field and balance computational efficiency, an atrous spatial pyramid pooling layer with dilated rates of $[1, 2, 4, 8]$ is utilized to obtain the further reconstructed features R_{s2}. The features are then concatenated and extracted to achieve the desired spatial resolution enhancement

$$R_{s2} = [r_1, r_2, r_4, r_8] = \mathcal{F}_{\text{ASPP}}(R_{s1}) \tag{9}$$

$$\mathcal{F}_{\text{MLR}}\left(\{F^i\}_{i=1}^4\right) = \mathcal{F}_{\text{PS}}(\mathcal{F}_{\text{LReLU}}(\mathcal{F}_{\text{CBAM}}(R_{s2}))). \tag{10}$$

C. Loss Function

To mitigate the impact of shadow areas in the RSI, we adopt a compromise approach that reduces the influence of pixels in shadow areas during network optimization. The overall loss function is defined as (11), where D_{GT} represents the ground truth DSM. To ensure stable training and reduce sensitivity to exceptional outliers, we normalize D_{GT} using the Z-score normalization [35]

$$L(\theta) = \frac{1}{N}\sum_{n=1}^{N}\left\|\left(D_{\text{SR}} - \widetilde{D}_{\text{GT}}\right)\cdot\omega(\text{Shadow})\right\|_1 \tag{11}$$

$$\widetilde{D}_{\text{GT}} = \frac{D_{\text{GT}} - \text{mean}(D_{\text{GT}})}{\text{std}(D_{\text{GT}})} \tag{12}$$

where $\|\sim\|_1$ represents the L_1 norm, and N is the total number of training samples.

We introduce pixel weights based on the shadow detection results of the RSI. The result is represented as S_{pred}, where each value $S_{\text{pred}}(i, j) \in (0, 1)$ indicates the likelihood of a pixel being a shadow. To achieve that, pixels with higher probabilities of being shadows have smaller weights during network optimization, we design a specific transformation using the following equation:

$$\omega(\text{Shadow}) = \exp(-kS_{\text{pred}}) + 1. \tag{13}$$

During our experiments, we utilize the shadow detection architecture borrowed from the SpA-Former [36] and pretrain it on the ISTD dataset [37]. In (13), we set the constant parameter k to 2.0. We will discuss the quality and impact of shadow detection in Section V-C.

IV. Experimental Settings

A. Datasets

US3D [38], [39], [40], [41] includes satellite images, airborne LiDAR covering approximately 100 km^2 over Jacksonville, Florida, Omaha, Nebraska, and Atlanta, Georgia. LiDAR was made publicly available by the U.S. Homeland Security Infrastructure Program, and the aggregate nominal pulse spacing is approximately 80 cm. The DSMs are derived from LIDAR data and share the same dimensions with corresponding RSIs. Each image has a size of 2048 × 2048, and the elevation data are centimeter-level. To train our model, all images are randomly cropped into 129 915 pairs with the size of 512 × 512, named us3d$_{\text{large}}$. These pairs are then split into training and validation sets in a 4:1 ratio. Furthermore, a smaller dataset us3d$_{\text{small}}$, is derived from us3d$_{\text{large}}$. It consists of 4779 training images and 597 testing images.

Besides, three unseen datasets are selected to evaluate the generalization capability. A detailed description is provided in Section V-D.

B. Downsampling Strategies

1) LiDAR-Mode Downsampling: LiDAR is a primary data source for obtaining DSMs in many practical applications. It offers high single-point accuracy but lacks correlation between adjacent points. To simulate this characteristic in a downsampling strategy, we propose a method called LiDAR-mode downsampling. Given a HR DSM $D_{\text{HR}} \in \mathbb{R}^{h\times w}$ and a downsampling factor s, we utilize nonoverlapping windows $W \in \mathbb{R}^{s\times s}$ and sample one pixel from each window to generate the $D_{\text{LR}} \in \mathbb{R}^{h/s\times w/s}$. During this process, we employ random sampling based on a Gaussian distribution within each window, which introduces a certain degree of randomness and disrupts the regularity of pixel arrangement. Unless otherwise specified, the experimental data is derived from LiDAR-mode downsampling method.

2) Stereo-Matching Downsampling: Another mainstream method for acquiring DSMs is involving triangulation through multiview RSIs. Compared to DSMs obtained by LiDAR, these photogrammetry-based DSMs exhibit stronger correlation among neighboring points but have significantly lower single-point accuracy. To simulate the desired characteristics, a blurring operator is employed on DSMs prior to downsampling. Specifically, given a HR DSM denoted as $D_{\text{HR}} \in \mathbb{R}^{h\times w}$, we generate a blurred DSM denoted as D_{blur} by reducing the resolution of D_{HR} using a bicubic interpolation and then upsampling it back to the initial resolution. Then, the LR DSM denoted as $D_{\text{LR}} \in \mathbb{R}^{h/s\times w/s}$ is generated by a nonoverlapping Gaussian kernel, which is defined as

$$\mathbb{K}_{i,j} = \frac{1}{2\pi\sigma^2}\exp\left(-\frac{(i-\frac{s}{2})^2 + (j-\frac{s}{2})^2}{2\sigma^2}\right)$$

where σ is randomly sampled from the range $[0.01, 4.0]$ and s represents the downsampling factor.

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TABLE I

QUANTITATIVE COMPARISON WITH SOTA SUPER-RESOLUTION ALGORITHMS ON THE $us3d_{large}$ DATASET. THE BEST PERFORMANCE IS SHOWN IN BOLD, AND THE DEPTH VALUES ARE MEASURED IN DECIMETERS

Down-sampling	Method	4×			8×			16×		
		MAE↓	RMSE↓	w-SSIM↑	MAE↓	RMSE↓	w-SSIM↑	MAE↓	RMSE↓	w-SSIM↑
LiDAR-mode	**Bicubic**	2.99	8.81	0.81	5.10	12.41	0.58	8.59	17.53	0.36
	GISR [29]	2.33	8.21	0.85	4.50	11.62	0.67	7.52	17.01	0.48
	TfaSR-v1 [28]	2.22	8.12	0.84	4.24	11.45	0.68	7.25	16.71	0.46
	BSRN [42]	2.00	7.95	0.88	3.94	11.41	0.71	6.88	16.34	0.52
	HAT [9]	1.89	7.76	0.89	3.82	11.25	0.74	6.45	15.87	0.56
	swinFIR [5]	1.88	7.74	0.89	3.82	11.27	0.74	6.37	15.78	0.57
	LIE-DSM(ours)	**1.81**	**7.64**	**0.90**	**3.38**	**10.61**	**0.76**	**5.18**	**13.54**	**0.65**
Stereo-matching	**Bicubic**	3.60	8.34	0.78	5.18	11.86	0.51	8.51	17.01	0.36
	GISR [29]	2.21	7.36	0.81	4.43	11.27	0.61	7.21	16.87	0.46
	TfaSR-v1 [28]	2.18	7.33	0.82	4.27	11.14	0.62	7.11	16.64	0.47
	BSRN [42]	2.04	7.24	0.84	3.98	11.03	0.62	6.99	16.24	0.47
	HAT [9]	1.91	6.89	0.87	3.78	10.56	0.66	6.57	15.70	0.53
	swinFIR [5]	1.87	6.84	0.87	3.75	10.47	0.67	6.41	15.58	0.54
	LIE-DSM(ours)	**1.78**	**6.59**	**0.89**	**3.28**	**9.84**	**0.74**	**4.87**	**13.32**	**0.65**

C. Train Details

The model is trained using a single NVIDIA RTX4090 GPU. During training, each sample is randomly cropped to a size of 384×384. Various online data augmentation techniques are applied to further augment the dataset, including random rotations, flips, and adjustments to the contrast and gamma values of the RSI within the range (0.8, 1.2). The models are trained for 200k iterations for each experiment with a batch size of 4. The Adam optimizer is employed with betas set to (0.9, 0.999) and an epsilon value of $1e{-}8$. The initial learning rate is set to $5e{-}4$ and is reduced every 50 000 iterations using a gamma value of 0.5.

V. EXPERIMENT

Evaluation Metrics: The mean absolute error (MAE) and root mean square error (RMSE) metrics are employed for objective evaluation metrics. However, considering the strong correlation between these metrics and $L1$ or $L2$ loss, the windowed structural similarity (w-SSIM) is also introduced, with window size of 11 in all experiments. The elevation values are measured in decimeters.

A. Comparison With Previous Methods

In this section, we compare our method with regular super-resolution methods using the $us3d_{large}$. We select SOTA networks [5], [9], [42] from the natural image domains, as most DSM methods are adapted from these networks. Furthermore, we compare our method with representative terrain super-resolution algorithms [28], [29] that have been specifically trained on the $us3d_{large}$ dataset. Table I demonstrates that our method significantly outperforms the other methods. Notably, as the downsampling rate increases, the impact of RSI becomes more pronounced, particularly under the 16× factor, where our method achieves improvements exceeding 15%. The results of LiDAR-mode and stereo-matching exhibit similar trends, but the latter achieves lower MAE and RMSE due to higher local correlation.

Fig. 3 showcases the results of the methods mentioned above under the 16× factor under LiDAR-mode downsampling. Even the best-performing super-resolution network fails to generate acceptable results under such as a large scaling factor commonly encountered in the DSM field. In contrast, our method effectively distinguishes neighboring buildings and accurately recovers the shape of the buildings. It shows exceptional performance even in case of many densely arranged small buildings. These results emphasize the importance of incorporating RSI information on the DSM super-resolution task.

Furthermore, we randomly select 1000 samples from the $us3d_{large}$ validation dataset and generate the RMSE distribution, shown in Fig. 4. Our method outperforms others with a smaller mean value and a higher concentration of values, and the RMSE metric exhibits greater concentration, indicating higher accuracy and improved reliability.

B. Network Validation

While there is a lack of comparable approaches in the field of DSM super-resolution, we choose guided depth map super-resolution (GDSR) as a suitable point of comparison, within the broader domain of multimodal fusion tasks. GDSR aims to improve the resolution of depth images associated with RGB images captured in general indoor and outdoor scenes. In the sense of the similarities in modal features between our task and GDSR, they can be comparable.

However, it is crucial to note that RSIs are often captured from a significant distance away from the ground, resulting in blurred depth relationships and higher object density. This phenomenon is particularly evident in densely populated urban areas. Additionally, DSM exhibits a more random distribution, with indoor and outdoor scenes generally adhering to specific patterns. For example, the center of the image typically has greater depth compared to the left and right sides, while the lower half tends to have smaller depths than the upper half. These inherent differences increase the complexity of the DSM super-resolution task and the challenge of effectively utilizing RSIs. By comparison, we can effectively demonstrate the effectiveness and distinctiveness of our network architecture for unique challenges on DSM super-resolution.

Specifically, six SOTA GDSR approaches CUNET [43], DKN [44], SVLRM [45], FDSR [46], AHMF [48], and DADA [47] are selected. To control variables, we train all models using the standard $L1$ loss on the $us3d_{small}$ dataset.

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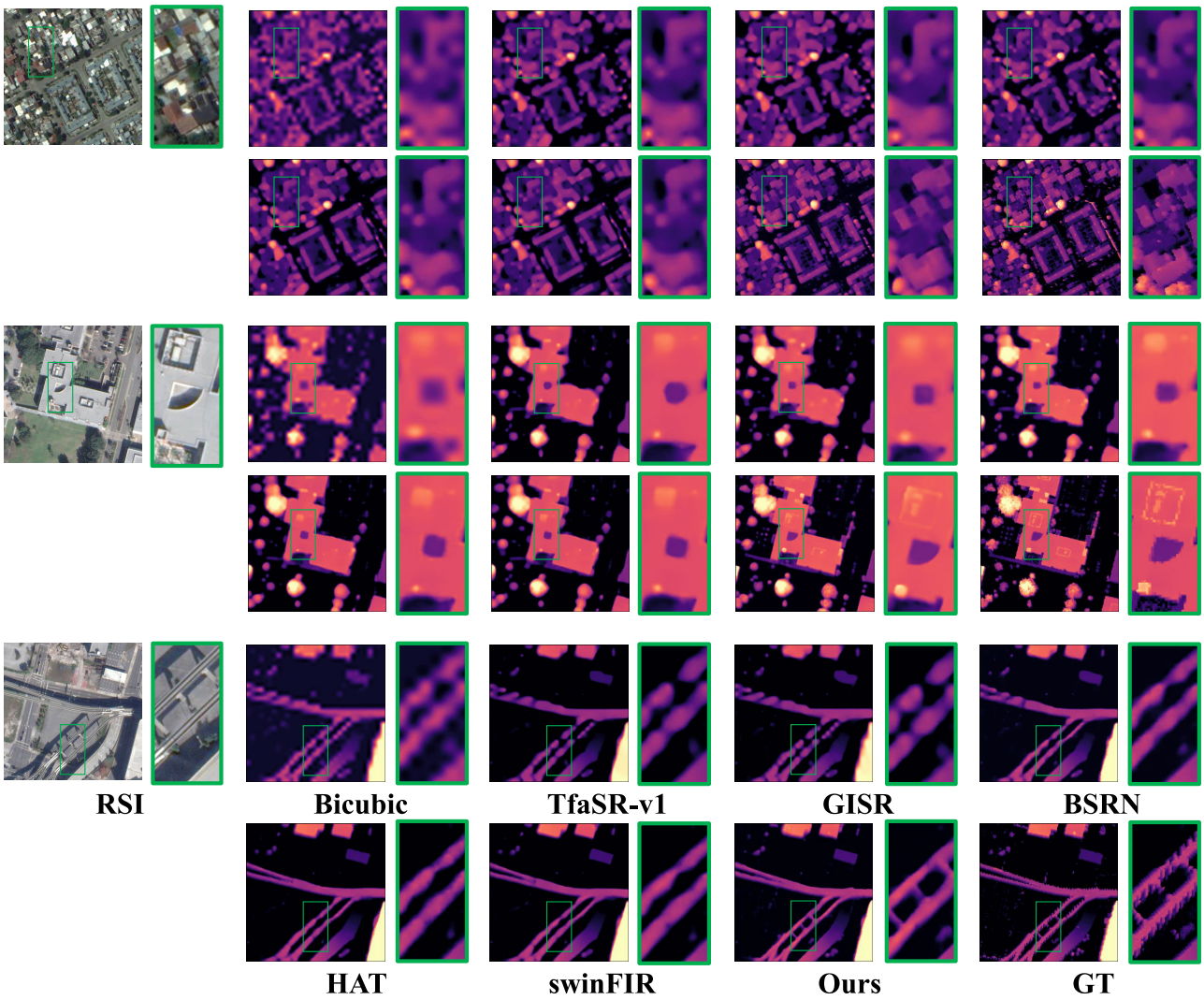


Fig. 3. Visual comparison with SOTA regular super-resolution algorithms under the 16× factor.

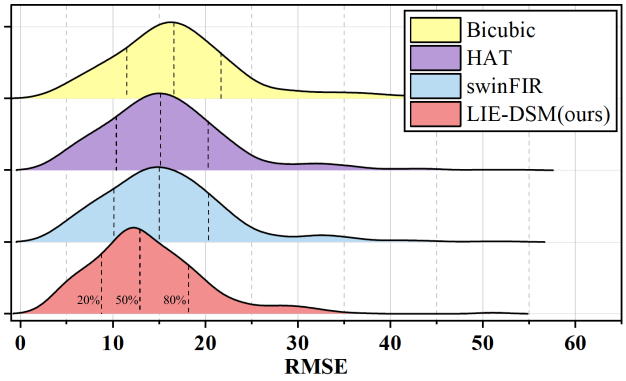


Fig. 4. RMSE distributions tested on $us3d_{large}$ dataset under the 16× factor.

The results in Table II clearly demonstrate the advantage of our method over other GDSR approaches. The visualization results in Fig. 5 show that our method exhibits remarkable similarity to GT in terms of shape recovery and data distribution. In contrast, even the best-performing GDSR method, AHMF, still exhibits a noticeable gap compared to our results, especially in complex scenes. This demonstrates that our network (LIE-DSMnet) is efficient in extracting high-frequency information from RSI.

1) Frequency Domain Analysis: To demonstrate the effectiveness in recovering high-frequency information, we computed the average Fast Fourier Transform of the results of Bicubic, AHMF, ours, and GT. Fig. 6 shows the amplitude difference between these results and GT, where white indicates a smaller amplitude difference, and the opposite is true in black. The center of the image represents the relatively low-frequency component, while the perimeter represents the relatively high-frequency component. With the same recovery of low-frequency components, our method exhibits significantly less difference from the GT in the high-frequency. It is due to the fact that our network is capable of recovering more high-frequency information, resulting in sharper edges and richer detail information.

2) Computational Complexity and Parameter Count: are also compared with GDSR method. The giga floating point operations per second (GFLOPS), the number of parameters (Params) and inference time are calculated based on an input size of 16 × 16, performing 8× super-resolution, and a batch size of 2. The results are presented in Fig. 7, where the reported inference time is the average of 1000 measurements. Notably, our network achieves high performance

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TABLE II

QUANTITATIVE COMPARISON WITH THE SOTA GDSR METHODS ON $us3d_{small}$ DATASET. THE BEST PERFORMANCE IS SHOWN IN BOLD, AND THE DEPTH VALUES ARE MEASURED IN DECIMETERS

Method	4×			8×			16×		
	MAE↓	RMSE↓	w-SSIM↑	MAE↓	RMSE↓	w-SSIM↑	MAE↓	RMSE↓	w-SSIM↑
CUNET [43]	2.31	8.48	0.87	4.40	12.17	0.68	7.45	16.98	0.45
DKN [44]	2.39	8.37	0.87	4.43	12.04	0.70	7.55	16.73	0.43
SVLRM [45]	2.24	8.53	0.86	4.31	12.15	0.69	7.32	16.84	0.48
FDSR [46]	2.12	8.35	0.88	4.16	11.86	0.72	6.90	16.26	0.52
DADA [47]	2.09	8.31	0.88	4.11	11.78	0.72	6.81	16.07	0.52
AHMF [48]	2.16	8.31	0.87	4.02	11.72	0.71	6.55	15.91	0.53
LIE-DSM(ours)	**1.95**	**8.01**	**0.90**	**3.83**	**11.27**	**0.76**	**6.07**	**15.02**	**0.57**

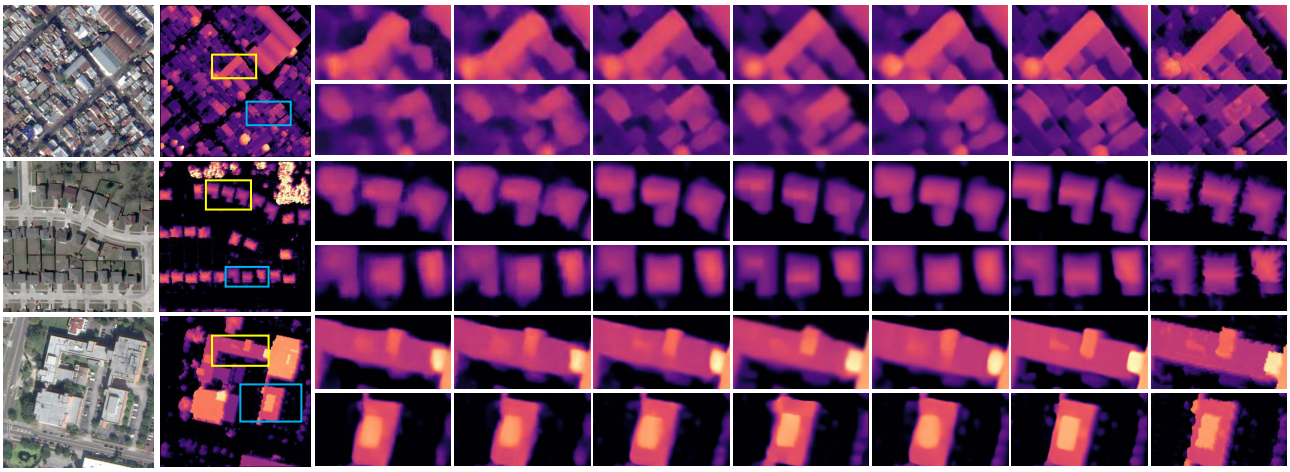


Fig. 5. Visual comparison with GDSR methods on $us3d_{large}$ dataset under the 16× factor. We magnify local regions.

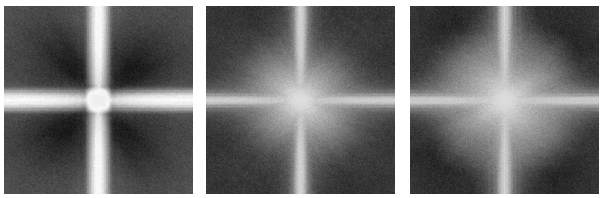


Fig. 6. Comparison of amplitude differences. (a) Bicubic. (b) AHMF. (c) LIE-DSMnet.

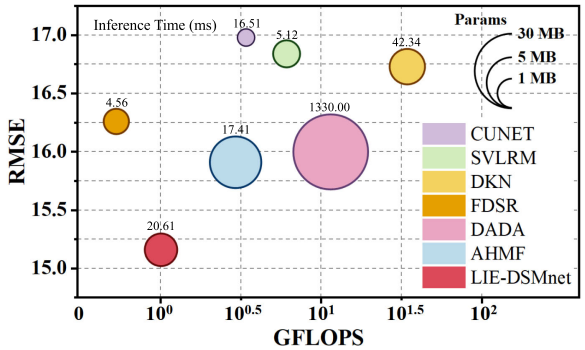


Fig. 7. Comparison with GDSR methods in aspects of GFLOPS and Params.

and demonstrates its lightweight and efficient characteristics, fulfilling the requirements for practical applications.

C. Ablation Study

We conduct experiments to evaluate two aspects of our algorithm: the efficiency of the shadow-weighted loss and the functional blocks within our network.

TABLE III

FUNCTION OF THE SHADOW-WEIGHTED LOSS (4×/8×/16×) UNDER THE 16× FACTOR ON THE $us3d_{large}$ DATASET, AND THE DEPTH VALUES ARE MEASURED IN DECIMETERS

	w/o SW	w/ SW
MAE↓	1.82 / 3.49 / 5.47	1.81 / 3.38 / 5.18
RMSE↓	7.66 / 10.73 / 13.92	7.64 / 10.61 / 13.54
w-SSIM↑	0.90 / 0.75 / 0.63	0.90 / 0.76 / 0.65

1) Shadow-Weighted Loss: To assess the effectiveness of the shadow weight on the loss function, we train our models under the 16× factor on the $us3d_{large}$ dataset using two different loss functions: shadow-weighted $L1$ loss ("w/ SW") and regular $L1$ loss ("w/o SW"). The quantitative results are shown in Table III. In addition, we visually analyze the results by selecting multiple local regions with prominent shadows, highlighting the disparities in super-resolution. In Fig. 8, the second row, labeled "Shadow" represents the pseudocolor of shadow probabilities, where probabilities increase from blue to red. We observe that "w/o SW" may contain errors due to the shadow area obscuring the true information. However, this deficiency can be partially mitigated by leveraging the shadow weight strategy.

2) Functional Blocks: Our network consists of three main functional blocks: GIHFF, DuFF, and MLR. To demonstrate the efficacy of these blocks, the model is trained on the $us3d_{small}$ dataset with regular $L1$ loss under the 16× factor on the following conditions: "full," "w/o GIHFF," "w/o

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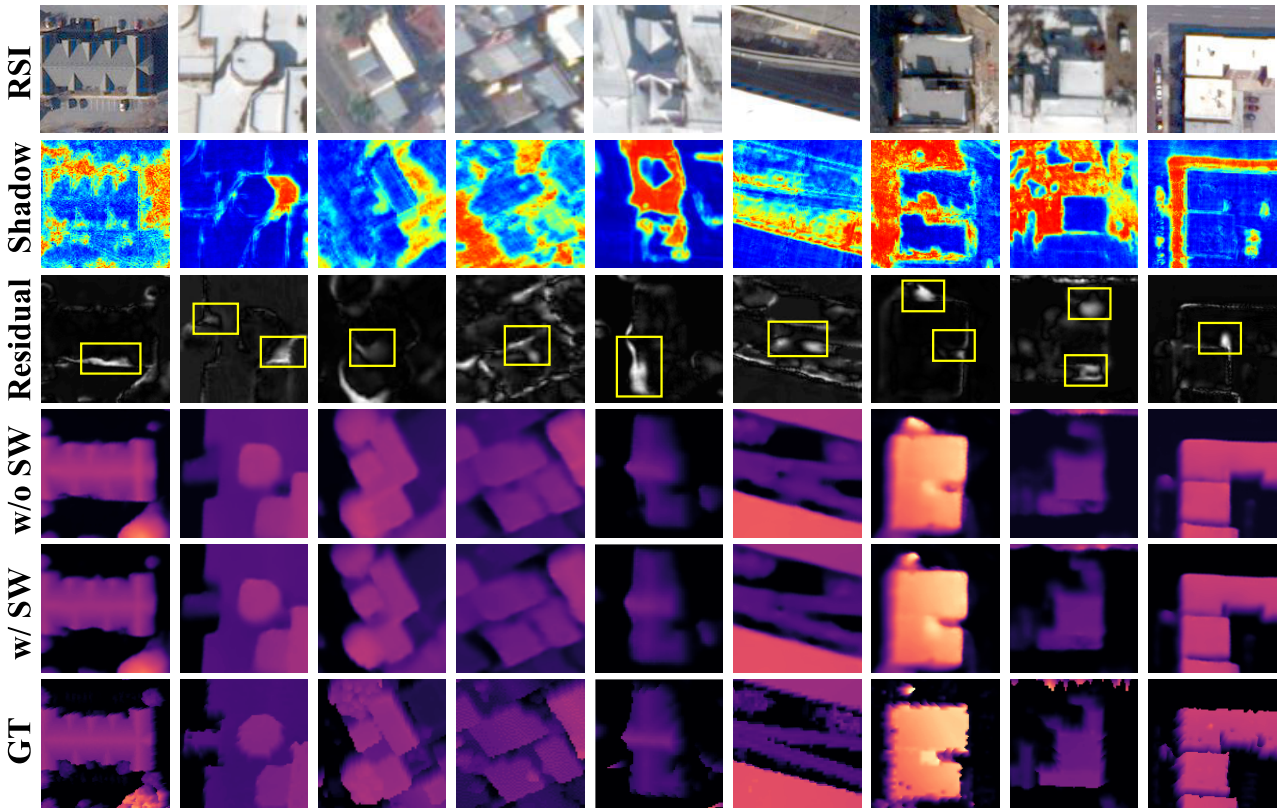


Fig. 8. Visual comparison of "w/ SW" and "w/o SW." The models are trained on us3d$_{\text{large}}$ dataset under the 16× factor. Please zoom the pdf to observe more details.

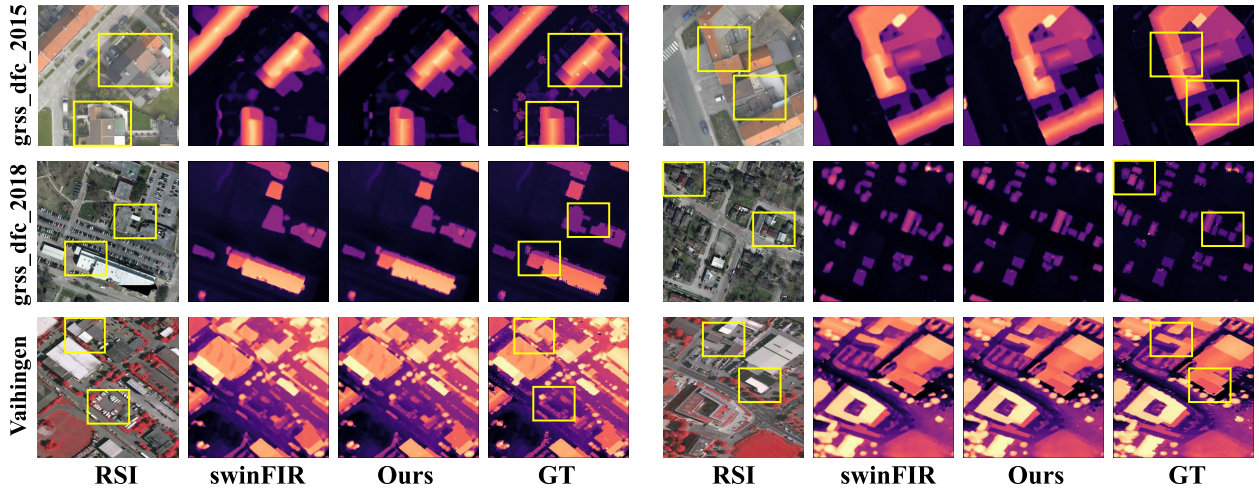


Fig. 9. Visual comparisons on three unseen datasets. All the models are trained on us3d$_{\text{large}}$ dataset. For grss_dfc_2015 [49], the super-resolution results are obtained under the 16× factor. For Vaihingen [50], [51], [52] and grss_dfc_2018 [53], the results are obtained under the 8× factor.

TABLE IV

CONTRIBUTIONS OF THREE FUNCTIONAL BLOCKS UNDER THE 16× FACTOR ON THE us3d$_{\text{small}}$ DATASET, AND THE DEPTH VALUES ARE MEASURED IN DECIMETERS

GIHFF	DuFF	MLP	MAE↓	RMSE↓	w-SSIM↑
	✓	✓	6.51	15.71	0.55
✓		✓	6.58	15.88	0.55
✓	✓		6.71	16.03	0.53
✓	✓	✓	6.07	15.02	0.57

DuFF," and "w/o MLR." For "w/o GIHFF," we use a regular convolutional layer instead of the GIHFF block without applying high-pass filtering. For "w/o DuFF," fused features are obtained by simply concatenating of the two sources. For "w/o MLR," we only utilize the final level fused features to reconstruct the results. The results depicted in Table IV demonstrate the effectiveness of the three blocks in achieving improved super-resolution performance.

D. Zero-Shot Experiment and More Visualization

To demonstrate the generalization capability of our method, we selected three unseen real test datasets. The swinFIR and ours models, trained on us3d$_{\text{large}}$ in Section V-A, are evaluated on the Vaihingen [50], [51], [52], grss_dfc_2015 [49] and

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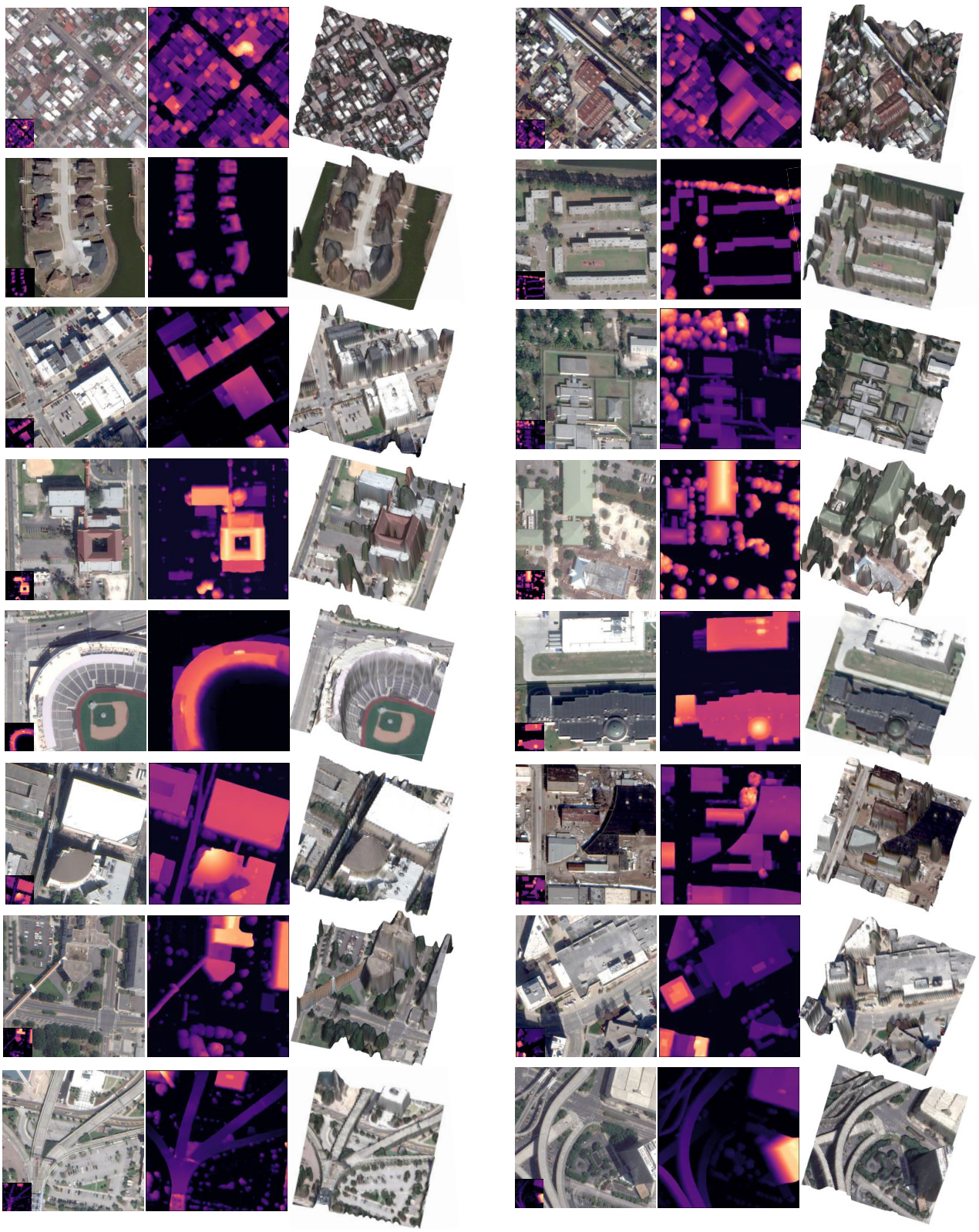


Fig. 10. Additional visualization results on us3d$_{\text{large}}$ dataset under the 16× factor.

grss_dfc_2018 [53] datasets. The details of the datasets are as follows.

1) *Vaihingen* includes true orthophoto and DSM, where the DSM was generated via dense image matching with Trimble INPHO 5.3 software. The ground sampling distance (GSD) of both is 9 cm. We randomly selected 27 scenes among all the scenes and resized them to 512 × 512 size.
2) *grss_dfc_2015* includes LiDAR data (at about 10 cm point spacing) and color orthophoto (at 5 cm spatial

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resolution), covering an urban and harbor area in Zeebruges, Belgium. We chose the scene "315 145–56 865," randomly cropped 41 test samples of 512 × 512 size.

3) *grss_dfc_2018* includes multispectral LiDAR point cloud data (at 0.5 m GSD) and RGB imagery (at 5 cm GSD). The LIDAR data are captured using the Optech Titam MW (14SEN/CON340), while the DiMAC ULTRALIGHT+ serves as a HR color imager. We randomly selected 31 patches, sized 512 × 512, from the entire scene.

The visualized results in Fig. 9 clearly demonstrate that the our method has good generalization ability to accommodate different unseen datasets. At the same time, our method consistently outperforms the other methods with more accurate and reliable details, especially in the case of buildings.

We provide additional visualization results on $us3d_{large}$ dataset under the 16× factor to demonstrate the performance of our LIE-DSM method in Fig. 10. For each visualization, we present the super-resolution DSM result alongside its corresponding 3-D model image. The 3-D model is generated by constructing a mesh using the DSM result and visualized using MeshLab, with the mesh points colored based on the RGB values of the corresponding pixel points. The super-resolution results exhibit sharp edges and smooth transition in surface depth, while the generated 3-D model accurately capture building contours and smooth building surfaces.

VI. Conclusion

This article introduces a novel algorithm called LIE-DSM that leverages the corresponding higher resolution RSI to enhance the resolution of DSM. The proposed method incorporates a network with self-designed functional modules and loss function that considers the unique characteristics of DSM and RSI. Experimental results demonstrate that our method achieves outstanding performance in both qualitative and quantitative assessments while maintaining outstanding speed and efficiency. Moreover, the super-resolution results achieved on three unseen datasets highlight the potential prospects for practical applications.

Limitations: Since our method is a supervised algorithm, one of the limitations is that the performance of our method depends on the quality of the HR DSM labels. Hence, in the next phase, our preliminary idea is to investigate an unsupervised or self-supervised approach for our LIE-DSM method. These approaches will allow building of a much larger dataset for training, further enhancing the algorithm's stability and generalization capability. Besides, considering that RSIs represent a mixture of various types of information, the geometric component is particular interest for DSM super-resolution. Hence, extracting the geometric information from RSIs before guiding the super-resolution process is a potential research direction.

References

[1] R. Bürgmann, P. A. Rosen, and E. J. Fielding, "Synthetic aperture radar interferometry to measure earth's surface topography and its deformation," *Annu. Rev. Earth Planet. Sci.*, vol. 28, no. 1, pp. 169–209, May 2000.

[2] G. Priestnall, J. Jaafar, and A. Duncan, "Extracting urban features from LiDAR digital surface models," *Comput., Environ. Urban Syst.*, vol. 24, no. 2, pp. 65–78, Mar. 2000.

[3] D. Zhu, X. Cheng, F. Zhang, X. Yao, Y. Gao, and Y. Liu, "Spatial interpolation using conditional generative adversarial neural networks," *Int. J. Geograph. Inf. Sci.*, vol. 34, no. 4, pp. 735–758, Apr. 2020.

[4] Z. Chen, X. Wang, Z. Xu, and W. Hou, "Convolutional neural network based DEM super resolution," *ISPRS Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 41, pp. 247–250, Jun. 2016.

[5] Y. Zhang et al., "NTIRE 2023 challenge on image super-resolution (X4): Methods and results," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2023, pp. 1864–1883.

[6] B. Lim, S. Son, H. Kim, S. Nah, and K. M. Lee, "Enhanced deep residual networks for single image super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 136–144.

[7] C. Ledig et al., "Photo-realistic single image super-resolution using a generative adversarial network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2017, pp. 4681–4690.

[8] J. Liang, J. Cao, G. Sun, K. Zhang, L. Van Gool, and R. Timofte, "SwinIR: Image restoration using Swin transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 1833–1844.

[9] X. Chen, X. Wang, J. Zhou, Y. Qiao, and C. Dong, "Activating more pixels in image super-resolution transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2023, pp. 22367–22377.

[10] Z. Xu, Z. Chen, W. Yi, Q. Gui, W. Hou, and M. Ding, "Deep gradient prior network for DEM super-resolution: Transfer learning from image to DEM," *ISPRS J. Photogramm. Remote Sens.*, vol. 150, pp. 80–90, Apr. 2019.

[11] A. Mohammed, M. Kashif, M. H. Zama, M. A. Ansari, and S. Ali, "Master GAN: Multiple attention is all you need: A multiple attention guided super resolution network for DEMs," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2023, pp. 5154–5157.

[12] S. Yao, Y. Cheng, F. Yang, and M. G. Mozerov, "A continuous digital elevation representation model for DEM super-resolution," *ISPRS J. Photogramm. Remote Sens.*, vol. 208, pp. 1–13, Feb. 2024.

[13] X. Li, H. Shen, R. Feng, J. Li, and L. Zhang, "DEM generation from contours and a low-resolution DEM," *ISPRS J. Photogramm. Remote Sens.*, vol. 134, pp. 135–147, Dec. 2017.

[14] M. Unser et al., "Fast B-spline transforms for continuous image representation and interpolation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 13, no. 3, pp. 277–285, Mar. 1991.

[15] X. Liu, H. Hu, and P. Hu, "Accuracy assessment of LiDAR-derived digital elevation models based on approximation theory," *Remote Sens.*, vol. 7, no. 6, pp. 7062–7079, May 2015.

[16] J. F. Kratzer, D. B. Hayes, and B. E. Thompson, "Methods for interpolating stream width, depth, and current velocity," *Ecolog. Model.*, vol. 196, nos. 1–2, pp. 256–264, 2006.

[17] Y. Zhang and W. Yu, "Comparison of DEM super-resolution methods based on interpolation and neural networks," *Sensors*, vol. 22, no. 3, p. 745, Jan. 2022.

[18] M. Karkee, B. L. Steward, and S. A. Aziz, "Improving quality of public domain digital elevation models through data fusion," *Biosyst. Eng.*, vol. 101, no. 3, pp. 293–305, 2008.

[19] H. Jhee, H.-C. Cho, H.-K. Kahng, and S. Cheung, "Multiscale quadtree model fusion with super-resolution for blocky artefact removal," *Remote Sens. Lett.*, vol. 4, no. 4, pp. 325–334, 2013.

[20] H. Jiang, L. Zhang, Y. Wang, and M. Liao, "Fusion of high-resolution DEMs derived from COSMO-SkyMed and TerraSAR-X InSAR datasets," *J. Geodesy*, vol. 88, pp. 587–599, Sep. 2014.

[21] Y. Tang, J. Zhang, L. Jing, and H. Li, "Digital elevation data fusion using multiple-point geostatistical simulation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 10, pp. 4922–4934, Oct. 2015.

[22] Z. Xu, X. Wang, Z. Chen, D. Xiong, M. Ding, and W. Hou, "Nonlocal similarity based dem super resolution," *ISPRS J. Photogramm. Remote Sens.*, vol. 110, pp. 48–54, Sep. 2015.

[23] D. Jiao, D. Wang, H. Lv, and Y. Peng, "Super-resolution reconstruction of a digital elevation model based on a deep residual network," *Open Geosci.*, vol. 12, no. 1, pp. 1369–1382, 2020.

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[24] O. Argudo, A. Chica, and C. Andujar, "Terrain super-resolution through aerial imagery and fully convolutional networks," *Comput. Graph. Forum*, vol. 37, no. 2, pp. 101–110, May 2018.

[25] A. Kubade, D. Patel, A. Sharma, and K. Rajan, "AFN: Attentional feedback network based 3D terrain super-resolution," in *Proc. Asian Conf. Comput. Vis.*, 2020, pp. 192–208.

[26] G. Chen, Y. Chen, J. P. Wilson, A. Zhou, Y. Chen, and H. Su, "An enhanced residual feature fusion network integrated with a terrain weight module for digital elevation model super-resolution," *Remote Sens.*, vol. 15, no. 4, p. 1038, 2023.

[27] R. Zhang, S. Bian, and H. Li, "RSPCN: Super-resolution of digital elevation model based on recursive sub-pixel convolutional neural networks," *ISPRS Int. J. Geo-Inf.*, vol. 10, no. 8, p. 501, 2021.

[28] Y. Zhang, W. Yu, and D. Zhu, "Terrain feature-aware deep learning network for digital elevation model superresolution," *ISPRS J. Photogramm. Remote Sens.*, vol. 189, pp. 143–162, Jul. 2022.

[29] X. Han, X. Ma, H. Li, and Z. Chen, "A global-information-constrained deep learning network for digital elevation model super-resolution," *Remote Sens.*, vol. 15, no. 2, p. 305, Jan. 2023.

[30] Y. Zhang, Z. Zheng, Y. Luo, Y. Zhang, J. Wu, and Z. Peng, "A CNN-based subpixel level DSM generation approach via single image super-resolution," *Photogramm. Eng. Remote Sens.*, vol. 85, no. 10, pp. 765–775, 2019.

[31] N. Imperatore and L. Dumas, "Contribution of super resolution to 3D reconstruction from pairs of satellite images," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 2, pp. 61–68, May 2022.

[32] M. Pashaei, M. J. Starek, H. Kamangir, and J. Berryhill, "Deep learning-based single image super-resolution: An investigation for dense scene reconstruction with UAS photogrammetry," *Remote Sens.*, vol. 12, no. 11, p. 1757, May 2020.

[33] Y. Chen et al., "Drop an Octave: Reducing spatial redundancy in convolutional neural networks with Octave convolution," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Jun. 2019, pp. 3435–3444.

[34] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. ECCV*, 2018, pp. 3–19.

[35] W. Yin et al., "Learning to recover 3D scene shape from a single image," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 204–213.

[36] X. Feng Zhang, C. Chen Gu, and S. Ying Zhu, "SpA-former: Transformer image shadow detection and removal via spatial attention," 2022, *arXiv:2206.10910*.

[37] J. Wang, X. Li, and J. Yang, "Stacked conditional generative adversarial networks for jointly learning shadow detection and shadow removal," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1788–1797.

[38] G. Christie, K. Foster, S. Hagstrom, G. D. Hager, and M. Z. Brown, "Single view geocentric pose in the wild," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2021, pp. 1162–1171.

[39] G. Christie, R. R. R. M. Abujder, K. Foster, S. Hagstrom, G. D. Hager, and M. Z. Brown, "Learning geocentric object pose in oblique monocular images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 14512–14520.

[40] B. Le Saux, N. Yokoya, R. Hänsch, and M. Brown, "2019 IEEE GRSS data fusion contest: Large-scale semantic 3D reconstruction," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 4, pp. 33–36, Jan. 2019.

[41] M. Bosch, K. Foster, G. Christie, S. Wang, G. D. Hager, and M. Brown, "Semantic stereo for incidental satellite images," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2019, pp. 1524–1532.

[42] Z. Li et al., "Blueprint separable residual network for efficient image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 833–843.

[43] X. Deng and P. L. Dragotti, "Deep convolutional neural network for multi-modal image restoration and fusion," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3333–3348, Oct. 2021.

[44] B. Kim, J. Ponce, and B. Ham, "Deformable kernel networks for joint image filtering," *Int. J. Comput. Vis.*, vol. 129, no. 2, pp. 579–600, Feb. 2021.

[45] J. Dong, J. Pan, J. S. Ren, L. Lin, J. Tang, and M.-H. Yang, "Learning spatially variant linear representation models for joint filtering," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 8355–8370, Nov. 2022.

[46] L. He et al., "Towards fast and accurate real-world depth super-resolution: Benchmark dataset and baseline," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 9229–9238.

[47] N. Metzger, R. C. Daudt, and K. Schindler, "Guided depth super-resolution by deep anisotropic diffusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 18237–18246.

[48] Z. Zhong, X. Liu, J. Jiang, D. Zhao, Z. Chen, and X. Ji, "High-resolution depth maps imaging via attention-based hierarchical multi-modal fusion," *IEEE Trans. Image Process.*, vol. 31, pp. 648–663, 2022.

[49] (Feb. 6, 2015). *2015 IEEE GRSS Data Fusion Contest*. [Online]. Available: http://www.grss-ieee.org/community/technical-committees/data-fusion

[50] M. Cramer, "The DGPF-test on digital airborne camera evaluation overview and test design," *Photogrammetrie Fernerkundung Geoinf.*, vol. 2010, pp. 73–82, May 2010.

[51] F. Rottensteiner et al., "THE ISPRS benchmark on urban object classification and 3D building reconstruction," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 3, pp. 293–298, Jul. 2012.

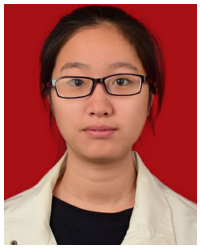
[52] I. Gerke, "Use of the stair vision library within the ISPRS 2D semantic labeling benchmark (Vaihingen)," ResearchGate, Dec. 2014. [Online]. Available: https://www.researchgate.net/publication/270104226_Use_of_the_Stair_Vision_Library_within_the_ISPRS_2D_Semantic_Labeling_Benchmark_Vaihingen?channel=doi&linkId=54ae59c50cf2828b29fcdf4b&showFulltext=true

[53] (Feb. 6, 2018). *2018 IEEE GRSS Data Fusion Contest*. [Online]. Available: http://www.grss-ieee.org/codmmunity/technical-committees/data-fusion



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